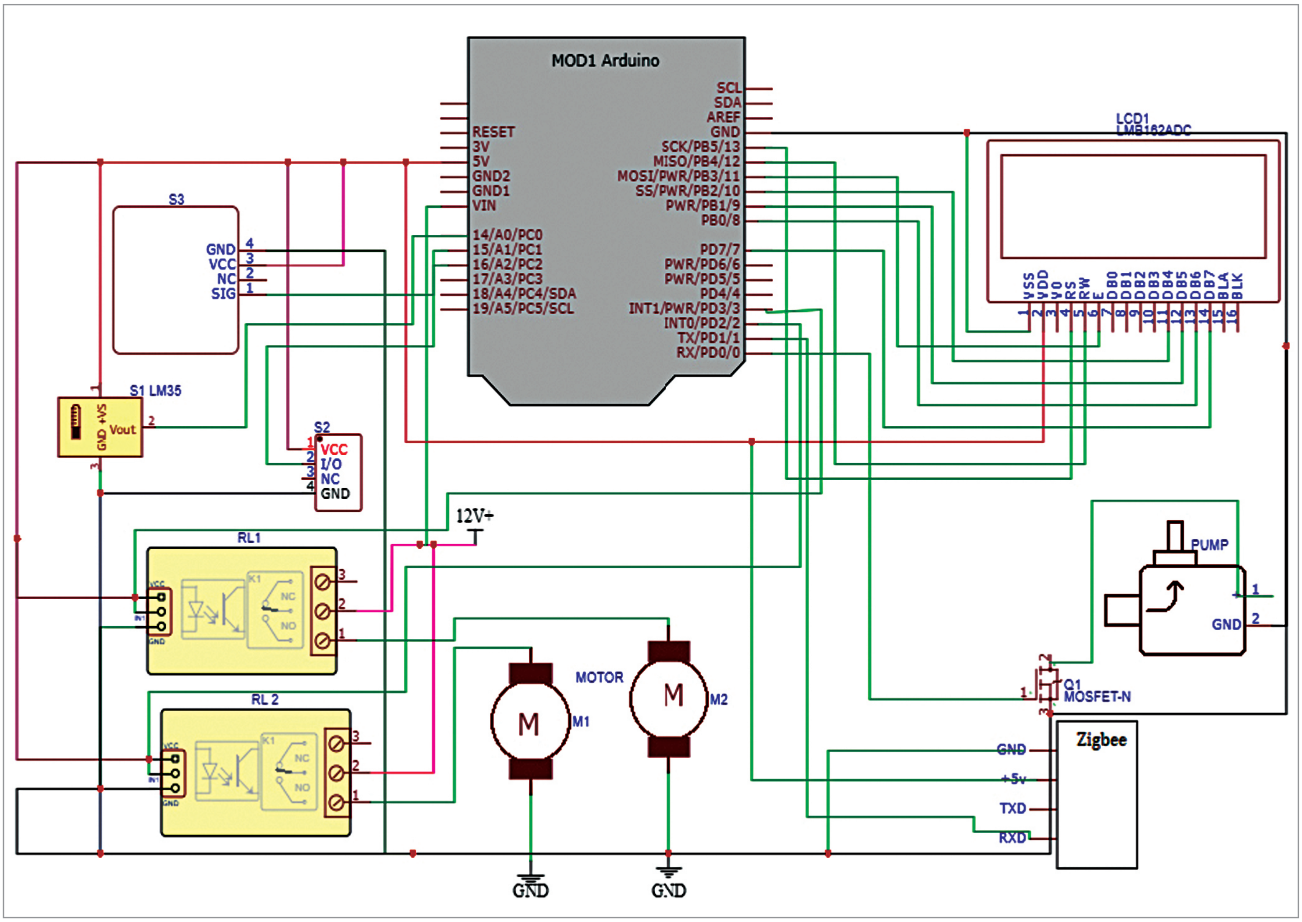

Fig. 3: Circuit diagram

BILL OF MATERIAL		
Components	**Quantity**	**Description**
Arduino Uno	1	Arduino is an open source electronics platform
Motor 5V	2	DC motor
Zigbee Tx and Rx	1 set	Zigbee Tx, Zigbee Rx
P55NF06 MOSFET	1	For pump driver
Humidity sensor	1	For humidity measurement
12V relay module (RL1, RL2)	2	Single changeover relay
16x2 LCD	1	LCD1
12V battery	1	For power supply
LM35 temperature sensor (S1)	1	For measuring temperature
Pump	1	For moving liquids/gases
MQ2 gas sensor (S2)	1	For gas sensing

relatively inexpensive compared to other microcontroller platforms and provide a simple, clear programming environment. Arduino boards are easy to handle and compatible with the others.

Zigbee. Zigbee is a wireless technology developed as an open global standard to address the unique needs of low-cost, low-power wireless IoT networks. The Zigbee standard operates on the IEEE 802.15.4 physical radio specification in unlicensed bands, including 2.4GHz, 900MHz, and 868MHz.

Temperature sensor. LM35 is a temperature sensor that outputs an analogue signal proportional to the instantaneous temperature. Its output voltage can be easily interpreted to obtain a temperature reading in degrees Celsius.

The advantage of LM35 over a thermistor is that it does not require any external calibration. The coating on it also protects it from self-heating. Low cost (approximately $0.95) and greater accuracy make it popular among hobbyists, DIY circuit makers, and students.

Gas Sensor. MQ2 is a commonly used gas sensor in MQ series. It is a metal oxide semiconductor (MOS) type gas sensor, also known as